

Energy poverty in the Netherlands

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Part 1: Identify a Social Problem

1.1 Describe the Social Problem

In recent years, energy prices in the Netherlands have increased significantly, leading to a substantial rise in fixed costs of Dutch households. This increase in fixed expenses reduces the financial resources available for those household, limiting their capacity to meet other essential needs. To effectively address this growing issue among those households, it is crucial to quantify and understand the extent to which rising energy prices drive poverty rates among the Dutch households.

Part 2: Data Sourcing

2.1 Load in the data

```
raw_energie <- read_csv2("Energietarieven.csv")
```

```
## i Using ',', ' as decimal and ''.' as grouping mark. Use 'read_delim()' for more control.
```

```
## Rows: 3 Columns: 11
```

```
## -- Column specification -----
```

```
## Delimiter: ";"
```

```
## dbl (9): Jaar, Aardgas transporttarief inc BTW Euro/jaar, Aardgas leverings ...
```

```
## lgl (2): Elektriciteit|Vast leveringstarief dynamisch inc btwEuro/jaar, Elek...
```

```
##
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
raw_inkomen <- read_csv2("Inkomen huishoudens samengevoegd.csv")
```

```
## i Using ',', ' as decimal and ''.' as grouping mark. Use 'read_delim()' for more control.
```

```
## Rows: 51 Columns: 8-- Column specification -----
```

```
## Delimiter: ";"
```

```
## chr (1): Regios
```

```
## dbl (7): Jaar, Particuliere huishoudens x 1 000, Particuliere huishoudens re...
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
dataset_samengevoegd <- raw_inkomen %>%
  left_join(raw_energie, by = "Jaar")
colnames(raw_energie)
```

```
## [1] "Jaar"
## [2] "Aardgas transporttarief inc BTW Euro/jaar"
## [3] "Aardgas leverings tarief vast inc btw Euro/jaar"
## [4] "Aardgas|Variabel leveringstarief inc btw Euro/m3"
## [5] "aardgas energiebelasting inc btw/m3"
## [6] "elektriciteit transporttarief inc btwEuro/jaar"
## [7] "elek vast leveringstarief inc btwEuro/jaar"
## [8] "Elektriciteit|Variabel leveringstarief contractprijs inc btwEuro/kWh"
## [9] "Elektriciteit|Vast leveringstarief dynamisch inc btwEuro/jaar"
## [10] "Elektriciteit|Variabel leveringstarief dynamisch inc btwEuro/kWh"
## [11] "Elektriciteit|Energiebelasting inc btwEuro/kWh"
```

```
colnames(raw_inkomen)
```

```
## [1] "Jaar"
## [2] "Regios"
## [3] "Particuliere huishoudens x 1 000"
## [4] "Particuliere huishoudens relatief%"
## [5] "Inkomen|Gemiddeld gestandaardiseerd inkomen 1 000 euro"
## [6] "Inkomen|Mediaan gestandaardiseerd inkomen 1 000 euro"
## [7] "Inkomen|Gemiddeld besteedbaar inkomen 1 000 euro"
## [8] "Inkomen|Mediaan besteedbaar inkomen 1 000 euro"
```

2.2 Short summary of the data sets

The data sets utilized in this study consist of official statistics sourced from Statistics Netherlands (CBS). The first data set contains average consumer energy tariffs, while the second data set comprises Dutch household income levels across the years 2021, 2022, and 2023, broken down by province.

A limitation of the first data set is that it does not capture the exact monthly or annual energy usage of individual households. However, it provides a reliable overview of the national average fixed and variable tariffs for both natural gas and electricity.

The second data set includes detailed information on standardized, median, and disposable household incomes. Utilizing these specific metrics ensures the most reliable and realistic representation of household financial capacity across the different Dutch provinces.

Part 3: Quantifying

Data cleaning

During the import of the Excel tables from the CBS website, it was found that the names of the columns did not match. The name of the first column in the Income datasheet was “jaar”, while the name of the first column in the Energie datasheet was “perioden”. That issue caused an error in R, it was solved by naming both first columns “jaar”. The energy data set was also cleaned by deleting the loose months and use the years instead. After these adjustments the data sets were merged by using `left_join()`, and R could read the files correctly.

Describing variabels

Creation of the new variables

```
# We berekenen de variabelen én hernoemen de oude kolommen in één stap!
dataset_berekend <- dataset_samengevoegd %>%
  mutate(
    # 1. We geven het inkomen meteen een kortere Engelse naam
    income_median = `Inkomen|Mediaan besteedbaar inkomen 1 000 euro`,

    # 2. Total gas costs calc (met de originele CBS-namen)
    gas_costs = `Aardgas transporttarief inc BTW Euro/jaar` +
      `Aardgas leverings tarief vast inc btw Euro/jaar` +
      (1200 * (`Aardgas|Variabel leveringstarief inc btw Euro/m3` + `aardgas energielasting`

    # 3. Total electricity costs calc (met de originele CBS-namen)
    electricity_costs = `elektriciteit transporttarief inc btwEuro/jaar` +
      `elek vast leveringstarief inc btwEuro/jaar` +
      (2500 * (`Elektriciteit|Variabel leveringstarief contractprijs inc btwEuro/kWh`

    # 4. Total energy costs combined
    total_energy_costs = gas_costs + electricity_costs,

    # 5. energy_burden (totale energiekosten)
    energy_burden = (total_energy_costs / (income_median * 1000)) * 100,

    # 6. gas_burden (pure gaskosten)
    gas_burden = (gas_costs / (income_median * 1000)) * 100,

    # 7. electricity_burden (pure stroomkosten)
    electricity_burden = (electricity_costs / (income_median * 1000)) * 100
  )
```

Explanation new variables

The new variables for this study are:

Energy_burden, represents the portion of the Dutch median disposable income that is spent on total energy costs (gas and electricity combined).

Gas_burden, represents the portion of the Dutch median disposable income that is spent on gas costs. This variable is calculated by adding the variable and fixed costs of gas based on a consumption of 1200m3 per year, the CBS guidelines show that 1200m3 is the normal consumption for a household per year.

Electricity_burden, represents the portion of the Dutch median disposable income that is spent on electricity costs. This variable is calculated by adding the variable and fixed costs of electricity based on a consumption of 2500kWh per year, the CBS guidelines show that 2500kWh is the normal consumption for a household per year(CBS,2023).

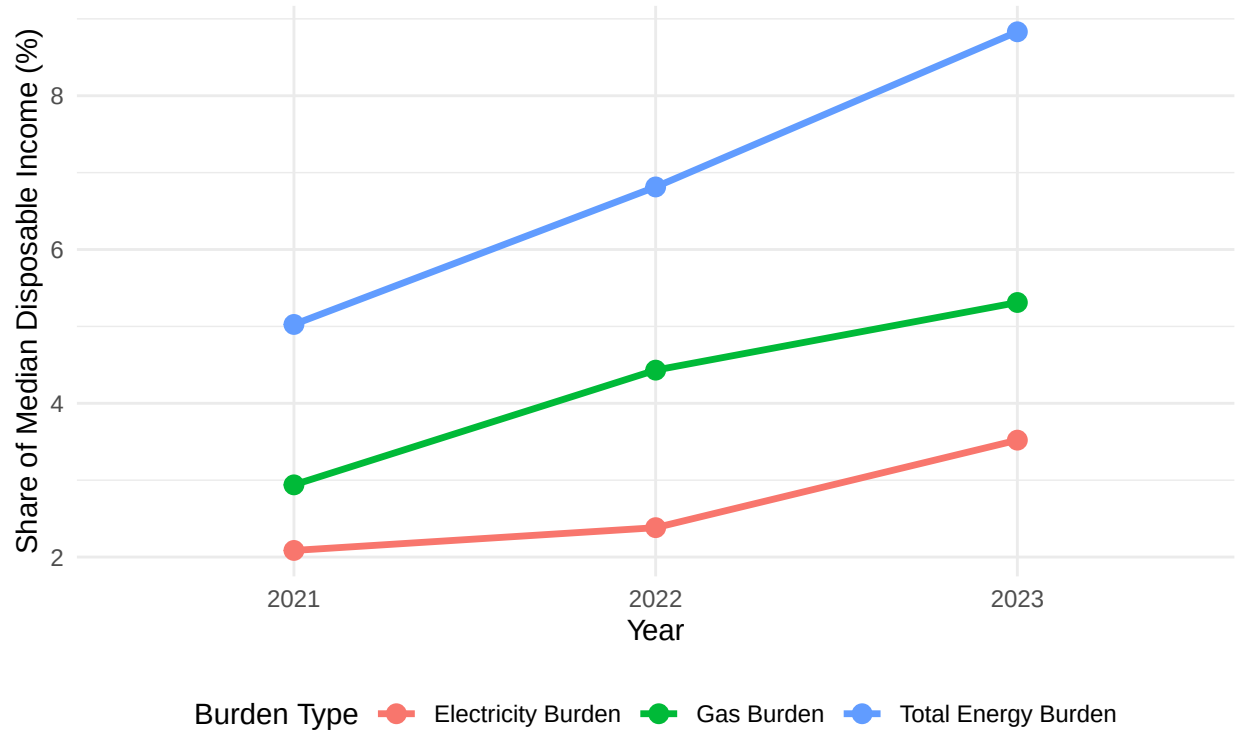
Data Visualisation

Figure 1

```
# 1. Data preperation
plot_data <- dataset_berekend %>%
  group_by(Jaar) %>%
  summarise(
    `Total Energy Burden` = mean(energy_burden, na.rm = TRUE),
    `Gas Burden` = mean(gas_burden, na.rm = TRUE),
    `Electricity Burden` = mean(electricity_burden, na.rm = TRUE)
  ) %>%
  pivot_longer(cols = ends_with("Burden"), names_to = "Burden_Type", values_to = "Percentage")

# 2. Plotting the Graph
ggplot(plot_data, aes(x = factor(Jaar), y = Percentage, group = Burden_Type, color = Burden_Type)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  labs(
    title = "Figure 1: Household Energy Burden Components (2021-2023)",
    subtitle = "National averages based on standard consumption",
    x = "Year",
    y = "Share of Median Disposable Income (%)",
    color = "Burden Type"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```

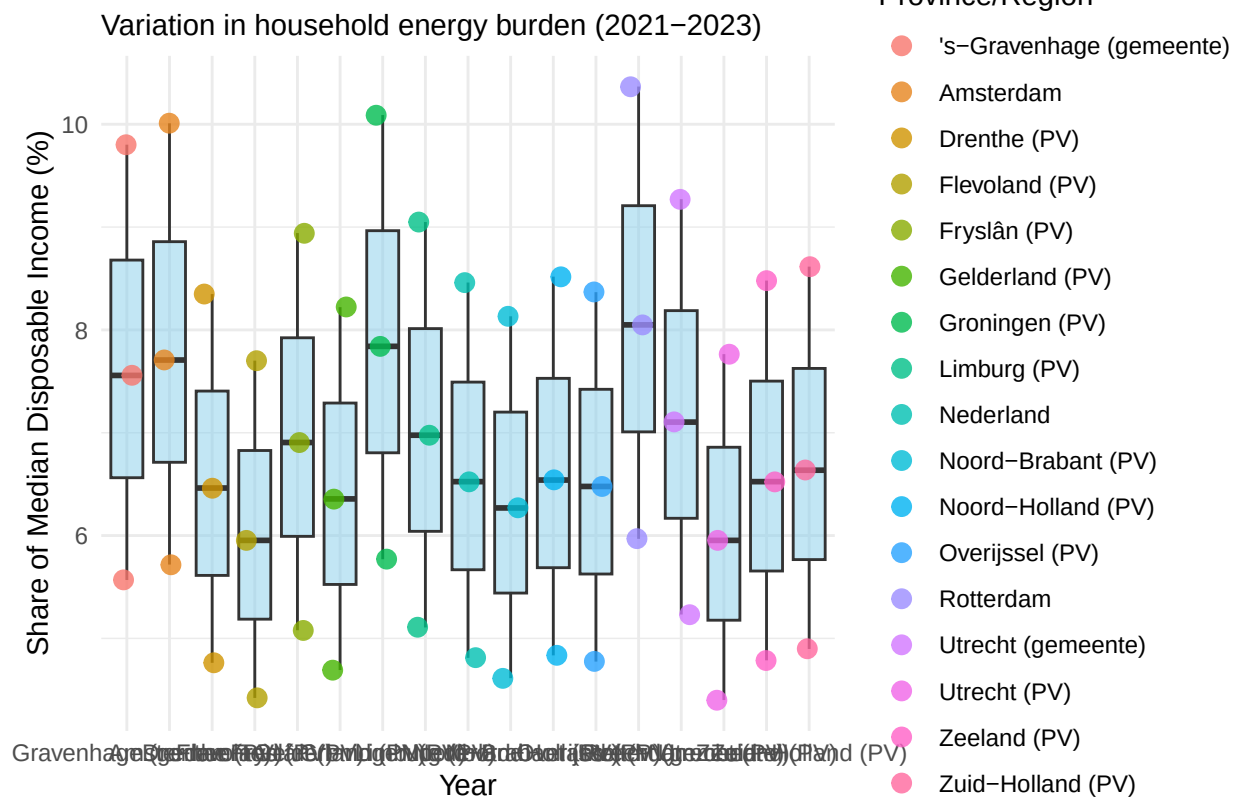
Figure 1: Household Energy Burden Components (2021–2023)
National averages based on standard consumption



#Figure 2

```
ggplot(dataset_berekend, aes(x = factor(Regios), y = energy_burden)) +
  geom_boxplot(fill = "skyblue", alpha = 0.5, outlier.shape = NA) +
  geom_jitter(width = 0.20, size = 3, aes(color = Regios), alpha = 0.8) +
  labs(
    title = "Figure 2: Distribution of Total Energy Burden Across Provinces",
    subtitle = "Variation in household energy burden (2021-2023)",
    x = "Year",
    y = "Share of Median Disposable Income (%)",
    color = "Province/Region"
  ) +
  theme_minimal() +
  theme(
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )
```

Figure 2: Distribution of Total Energy Burden Across Provinces



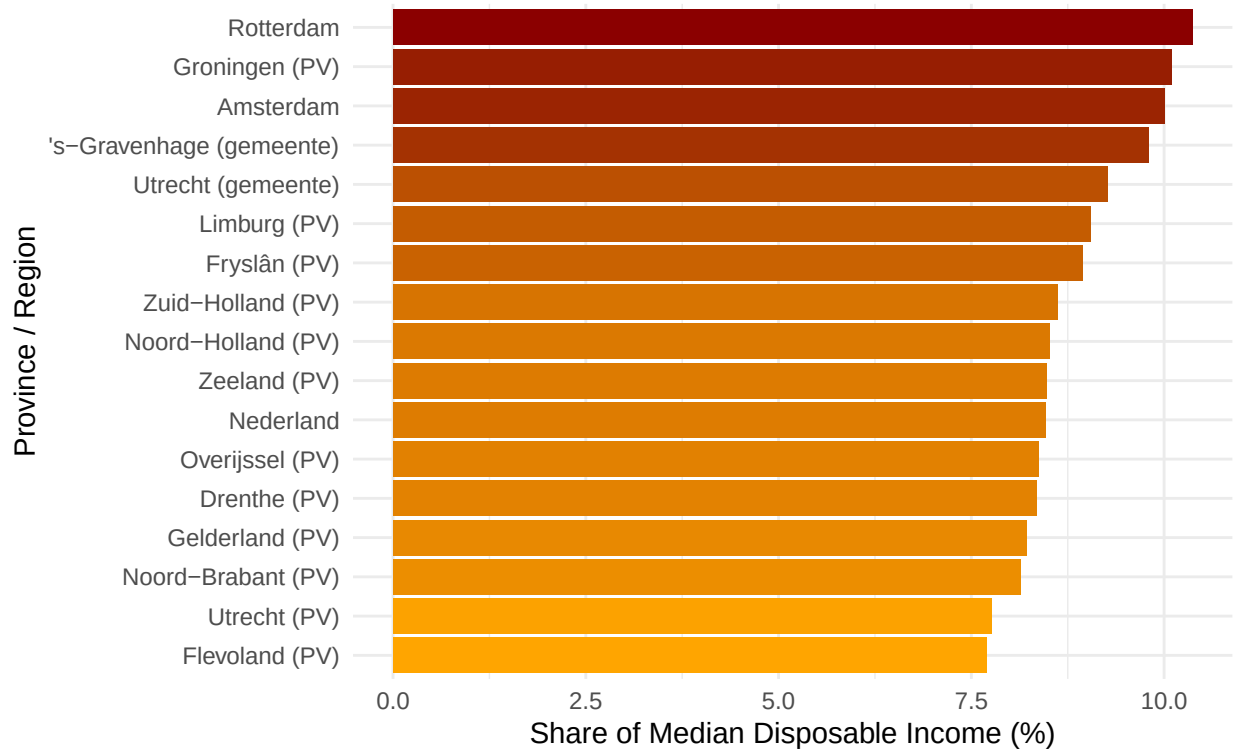
#Figure 3

```
data_2023 <- dataset_berekend %>%
  filter(Jaar == 2023)

ggplot(data_2023, aes(x = reorder(Regios, energy_burden), y = energy_burden, fill = energy_burden)) +
  geom_col() +
  coord_flip() + # Draait de grafiek om zodat de provincienamen leesbaar zijn
  scale_fill_gradient(low = "orange", high = "darkred") +
  labs(
    title = "Figure 3: Total Energy Burden by Province in 2023",
    subtitle = "Provinces sorted from lowest to highest energy burden",
    x = "Province / Region",
    y = "Share of Median Disposable Income (%)",
    fill = "Burden (%)"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold")
  )
```

Figure 3: Total Energy Burden by Province in 2023

Provinces sorted from lowest to highest energy burden

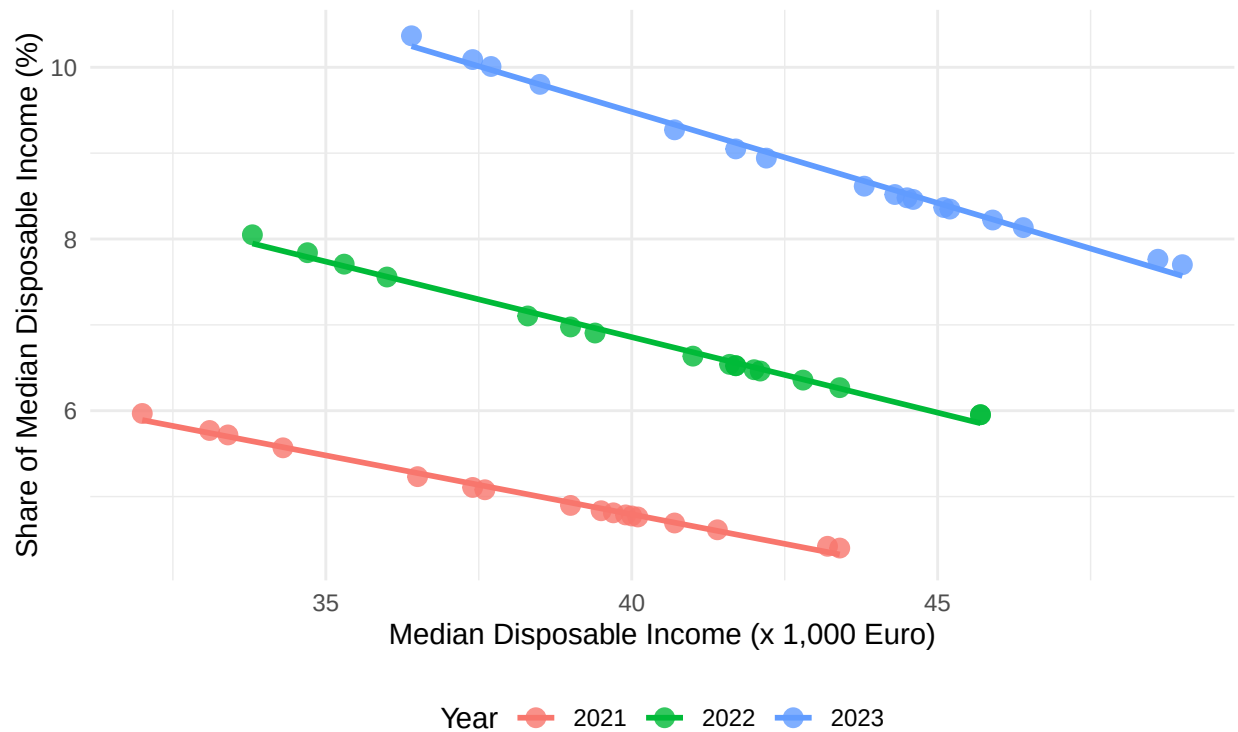


```
ggplot(dataset_berekend, aes(x = income_median, y = energy_burden, color = factor(Jaar))) +
  geom_point(size = 3, alpha = 0.8) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1) + # Voegt een trendlijn toe
  labs(
    title = "Figure 4: Correlation Between Median Income and Energy Burden",
    subtitle = "Each point represents a province in a specific year",
    x = "Median Disposable Income (x 1,000 Euro)",
    y = "Share of Median Disposable Income (%)",
    color = "Year"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "bottom"
  )
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Figure 4: Correlation Between Median Income and Energy Burden

Each point represents a province in a specific year



Discussion

Our findings state that, within the limitations mentioned earlier, the share of the Dutch median income spent on energy bills increased significantly each year between 2021 and 2023. It is important to note that precise figures are difficult to determine. However, the fact remains that standard disposable income has not increased as rapidly as energy costs. This trend highlights the growing importance of policies aimed at improving energy affordability and protecting vulnerable households in the Netherlands.